Department of Computer Science and Engineering, UCE(A), OU
Osmania University
B.E. (CSE) IV-Semester I-Internal Examination-2025

Subject Name: OPERATING SYSTEMS

Time: 12:00 to 1: 30 PM

Date: 28-03-2025
Max. Marks: 30

Note: Answer all Question in Part-A and any two in Part-B

Q.NO	Part –B (5*2 = 10 Marks)	Marks	Blooms Taxonomy Level	Course Outcome
1	Differentiate between preemptive and non-preemptive scheduling	2.0	3	2
2	What is a Process Control Block (PCB)?	2.0	2	1
3	Consider the following code fragment: **if** (fork() == 0) { a = a + 5; printf("%d,%d\\n", a, &a); } **else** { a = a –5; printf("%d, %d\\n", a, &a); } Let u, v be the values printed by the parent process, and x, y be the values printed by the child process. Which one of the following is TRUE? a. u = x + 10 and v = y b. u = x + 10 and v != y c. u + 10 = x and v = y d. u + 10 = x and v != y	2.0	2	2
4	The following C program is executed on a Unix/Linux system: #include<unistd.h> int main() { int i; for(i=0;i<10;i++) if (i%2==0) fork (); return 0; } The total number of child processes created is _____.	2.0	1	1
5	OS performs the following actions when a new process is created: (i) Allocates the memory and other resources to the process. (ii) Assigns process id and priority (iii) Creates a process control block (PCB) for the process (iv) Sets up the process environment (v) Initializes resource accounting information for the process. What would be the correct sequence of the above actions? (A) 4, 3, 1, 2, 5 (B) 3, 2, 1, 4, 5 (C) 4, 3, 5, 2, 1 (D) 3, 4, 2, 5, 1	2.0	3	2

Part –B (2*10 = 20 Marks)				
6 (a)	What is the relationship between operating systems and computer hardware? Distinguish between multiprocessing and multiprogramming?	5.0	2	1
(b)	Consider the set of jobs < J1, J2, J3, J4, J5 > assumed to arrive at time sequence as < 0,1,2,6,12> with the length of the CPU processing time < 7, 5, 3, 2, 3>. (i) Draw the Giant Chart illustrating their execution using Shortest Job First (SJF) and Shortest Renaming Time First (SRTF / SJF Preemptive) scheduling Algorithms. (ii)Calculate the Average Turn Around time and Average waiting time for both above scheduling algorithms.	5.0	2	2
7 (a)	What is meant by scheduling? Explain Round robin scheduling. Give an Example.	5.0	1	2
(b)	Describe the First fit, Best fit and Worst fit allocation algorithms. Given memory partitions of 100K, 500K, 200K, 300K, and 600K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes 212K, 417K, 112K, and 426K (in order)? Which algorithm makes the most efficient use of memory?	5.0	1	2
8 (a)	Define Deadlock. State the four necessary conditions for a deadlock situation to occur in a system?	5.0	2	2
(b)	Let the ***duration of a time slice be 4 msec***, which is to say CPU switches between processes every 4 msec in a round-robin fashion. Find Average waiting time, Average turnaround time and Throughput.	5.0	2	2

Process	P1	P2	P3
Burst time (msec)	24	3	3

